

DDR4-UDIMM routing guide

16 GB DDR4-3200 UDIMM, Raw Card A3. Which signal is routed on which layer, by component designator. Tables generated from DDR4-UDIMM — 16 GB DDR4.PcbDoc; rules from JESD21-C 4.20.26 (main spec Tables 10-14 and Annex A).

1. Layer stack

Layer	Name	Type	References	What is routed here
L1	Top Layer 1	signal, microstrip	L2 (plane, 70 um)	DQ / DQS / DM escapes, finger stubs, short address hops
L2	L2_PWR_GND	plane	-	reference for L1 and L3; GND under the data groups
L3	L3_DQ_ADDR	signal, stripline	L2 (plane, 80 um)	DQ / DQS / DM DRAM side; address permitted
L4	L4_VDD	plane	-	reference for L5; VDD
L5	L5_ADDR	signal, stripline	L4 (VDD, 80 um)	address and command only
L6	L6_ADDR_CK	signal, stripline	L7 (plane, 80 um)	address, command and the clock pairs
L7	L7_PWR_GND	plane	-	reference for L6 and L8
L8	Bottom Layer 1	signal, microstrip	L7 (plane, 70 um)	DQ / DQS / DM DRAM side, clocks

Never route on L2, L4 or L7. They are solid planes and they are the return path for everything else; a track on a plane cuts that return path.

2. Layer assignment rules (enforced by DRC)

Signal group	Allowed layers	Altium rule	Why
DQ / DQS / DM	L1, L3, L8 only	RoutingLayers_DATA	L5 and L6 reference VDD; data return current needs GND (Table 12)
CK0_T / CK0_C	L1, L6, L8 only	RoutingLayers_CK	clock references VDD or a plane, per Table 12
Address, control, RESET_n, ALERT_n	L1, L3, L5, L6, L8	RoutingLayers_ADDR	address references VDD
Everything else	L1, L3, L5, L6, L8	RoutingLayers	L2, L4 and L7 are planes - never route on them

3. Layer by layer

L1 - Top Layer 1

signal (microstrip, references L2)

Signal group	Components	From - to	Width / impedance
Data, all 8 bytes	R2-R12, R14-R24, ... R86-R96 (88 resistors)	J1 finger to resistor pad 2, and resistor pad 1 to the DRAM fan-out via	0.10 mm, 50 ohm
Address and command	R97-R126 to J1 and the DRAMs	finger stubs and short hops	0.075 mm, 55 ohm
ZQ	R1, R13, R25, R37, R49, R61, R73, R85	DRAM ZQ ball to 240 ohm to GND	0.10 mm
ALERT_n	R102	47 ohm pull-up to VDD, before the first DRAM (U1)	0.075 mm

L3 - L3_DQ_ADDR

signal (stripline, references L2 = GND)

Signal group	Components	From - to	Width / impedance
Data, DRAM side	NetR2_1 ... NetR96_1	resistor pad 1 to the DRAM ball, via L1	0.10 mm, 50 ohm
Address and command	any of A0-A16, BA, BG, ACT, CKE, CS, ODT, PAR, WE	permitted but keep it clear of the byte lanes	0.075 mm, 55 ohm

L5 - L5_ADDR

signal (stripline, references L4 = VDD)

Signal group	Components	From - to	Width / impedance
Address	A0-A13, A10_AP, A12_BC, A15_CAS, A16_RAS	J1 to U1 -> U2 -> ... -> U8 fly-by	0.075 mm, 55 ohm
Bank / command	BA0, BA1, BG0, BG1, ACT, WE, PAR	same fly-by	0.075 mm, 55 ohm
Control	CKE, CS, ODT, RESET_n	same fly-by	0.075 mm, 55 ohm

L6 - L6_ADDR_CK

signal (stripline, references L7)

Signal group	Components	From - to	Width / impedance
Clock pair	CK0_T / CK0_C	J1 to every DRAM, fly-by, terminated at R115 / R117 and C56	0.075 / 0.10 mm gap, 93 ohm differential
Address and command overflow	whatever does not fit on L5	same fly-by	0.075 mm, 55 ohm

L8 - Bottom Layer 1

signal (microstrip, references L7)

Signal group	Components	From - to	Width / impedance
Data, DRAM side	NetR2_1 ... NetR96_1	second choice when L3 is congested	0.10 mm, 50 ohm
Clocks	CK0_T / CK0_C	permitted	93 ohm differential

4. Data byte lanes, by designator

Every data line is J1 finger -> Rn pad 2 -> Rn pad 1 (15 ohm) -> DRAM ball. Route pad 1 to the ball on L1 + L3 (or L1 + L8). The target is the byte's longest line; each line must land within 1.0 mm of it. **Route the whole byte before tuning any of it:** the target below is only the longest line routed *so far*, so in a byte that is still mostly unrouted it will rise. A DRAM-side length of about 0.57 mm means nothing is routed yet - that is just the ball-to-via stub.

Byte 0 - U1 (target 19.56 mm, window 18.56 to 20.56)

Resistor	Finger net	J1 pin	DRAM-side net	DRAM ball	Layers	finger mm	DRAM mm	full mm	to do
R2	DQ0	5	NetR2_1	U1-C2	L1 + L3 or L1 + L8	2.95	9.43	13.38	+6.19 mm
R3	DQ1	150	NetR3_1	U1-B7	L1 + L3 or L1 + L8	5.45	4.00	10.45	+9.11 mm
R4	DQ2	12	NetR4_1	U1-D3	L1 + L3 or L1 + L8	2.95	8.27	12.22	+7.34 mm
R5	DQ3	157	NetR5_1	U1-D7	L1 + L3 or L1 + L8	5.45	11.21	17.66	+1.91 mm
R6	DQ4	3	NetR6_1	U1-D2	L1 + L3 or L1 + L8	2.95	11.02	14.97	+4.59 mm
R7	DQ6	10	NetR7_1	U1-E3	L1 + L3 or L1 + L8	2.95	12.06	16.01	+3.56 mm
R8	DQ7	155	NetR8_1	U1-E7	L1 + L3 or L1 + L8	5.45	13.11	19.56	in window
R9	DQS0_T	153	NetR9_1	U1-C3	L1 + L3 or L1 + L8	5.54	7.01	13.55	+6.01 mm
R10	DQS0_C	152	NetR10_1	U1-B3	L1 + L3 or L1 + L8	5.54	6.68	13.22	+6.34 mm
R11	DM0	7	NetR11_1	U1-A7	L1 + L3 or L1 + L8	2.95	6.00	9.95	+9.61 mm
R12	DQ5	148	NetR12_1	U1-D8	L1 + L3 or L1 + L8	5.45	5.92	12.37	+7.20 mm

Byte 1 - U2 (target 14.78 mm, window 13.78 to 15.78)

Resistor	Finger net	J1 pin	DRAM-side net	DRAM ball	Layers	finger mm	DRAM mm	full mm	to do
R14	DQ8	16	NetR14_1	U2-C2	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R15	DQ9	161	NetR15_1	U2-B7	L1 + L3 or L1 + L8	5.47	0.74	7.21	route it
R16	DQ10	23	NetR16_1	U2-D3	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R17	DQ11	168	NetR17_1	U2-D7	L1 + L3 or L1 + L8	5.45	0.57	7.02	route it
R18	DQ12	14	NetR18_1	U2-D2	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R19	DQ14	21	NetR19_1	U2-E3	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R20	DQ15	166	NetR20_1	U2-E7	L1 + L3 or L1 + L8	5.45	0.57	7.02	route it
R21	DQS1_T	164	NetR21_1	U2-C3	L1 + L3 or L1 + L8	5.54	8.24	14.78	in window

R22	DQS1_C	163	NetR22_1	U2-B3	L1 + L3 or L1 + L8	5.54	8.24	14.78	in window
R23	DM1	18	NetR23_1	U2-A7	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R24	DQ13	159	NetR24_1	U2-D8	L1 + L3 or L1 + L8	5.45	7.95	14.40	in window

Byte 2 - U3 (target 16.43 mm, window 15.43 to 17.43)

Resistor	Finger net	J1 pin	DRAM-side net	DRAM ball	Layers	finger mm	DRAM mm	full mm	to do
R26	DQ16	27	NetR26_1	U3-C2	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R27	DQ17	172	NetR27_1	U3-B7	L1 + L3 or L1 + L8	5.47	0.57	7.04	route it
R28	DQ18	34	NetR28_1	U3-D3	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R29	DQ19	179	NetR29_1	U3-D7	L1 + L3 or L1 + L8	5.45	0.57	7.02	route it
R30	DQ20	25	NetR30_1	U3-D2	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R31	DQ22	32	NetR31_1	U3-E3	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R32	DQ23	177	NetR32_1	U3-E7	L1 + L3 or L1 + L8	5.45	0.57	7.02	route it
R33	DQS2_T	175	NetR33_1	U3-C3	L1 + L3 or L1 + L8	5.56	8.12	14.69	+1.74 mm
R34	DQS2_C	174	NetR34_1	U3-B3	L1 + L3 or L1 + L8	5.56	9.87	16.43	in window
R35	DM2	29	NetR35_1	U3-A7	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R36	DQ21	170	NetR36_1	U3-D8	L1 + L3 or L1 + L8	5.45	0.57	7.02	route it

Byte 3 - U4 (target 17.33 mm, window 16.33 to 18.33)

Resistor	Finger net	J1 pin	DRAM-side net	DRAM ball	Layers	finger mm	DRAM mm	full mm	to do
R38	DQ24	38	NetR38_1	U4-C2	L1 + L3 or L1 + L8	3.01	0.57	4.58	route it
R39	DQ25	183	NetR39_1	U4-B7	L1 + L3 or L1 + L8	5.53	0.57	7.10	route it
R40	DQ26	45	NetR40_1	U4-D3	L1 + L3 or L1 + L8	3.16	0.57	4.72	route it
R41	DQ27	190	NetR41_1	U4-D7	L1 + L3 or L1 + L8	5.68	0.57	7.24	route it
R42	DQ28	36	NetR42_1	U4-D2	L1 + L3 or L1 + L8	2.97	0.57	4.54	route it
R43	DQ30	43	NetR43_1	U4-E3	L1 + L3 or L1 + L8	3.12	0.57	4.68	route it
R44	DQ31	188	NetR44_1	U4-E7	L1 + L3 or L1 + L8	5.64	0.57	7.20	route it
R45	DQS3_T	186	NetR45_1	U4-C3	L1 + L3 or L1 + L8	5.71	10.62	17.33	in window
R46	DQS3_C	185	NetR46_1	U4-B3	L1 + L3 or L1 + L8	5.67	10.62	17.29	in window
R47	DM3	40	NetR47_1	U4-A7	L1 + L3 or L1 + L8	3.05	0.57	4.62	route it
R48	DQ29	181	NetR48_1	U4-D8	L1 + L3 or L1 + L8	5.49	0.57	7.06	route it

Byte 4 - U5 (target 14.73 mm, window 13.73 to 15.73)

Resistor	Finger net	J1 pin	DRAM-side net	DRAM ball	Layers	finger mm	DRAM mm	full mm	to do
R50	DQ32	97	NetR50_1	U5-C2	L1 + L3 or L1 + L8	3.20	0.57	4.77	route it
R51	DQ33	242	NetR51_1	U5-B7	L1 + L3 or L1 + L8	5.70	0.57	7.27	route it
R52	DQ34	104	NetR52_1	U5-D3	L1 + L3 or L1 + L8	3.20	0.57	4.77	route it
R53	DQ35	249	NetR53_1	U5-D7	L1 + L3 or L1 + L8	5.70	0.57	7.27	route it
R54	DQ36	95	NetR54_1	U5-D2	L1 + L3 or L1 + L8	3.20	0.57	4.77	route it
R55	DQ38	102	NetR55_1	U5-E3	L1 + L3 or L1 + L8	3.20	0.57	4.77	route it
R56	DQ39	247	NetR56_1	U5-E7	L1 + L3 or L1 + L8	5.70	0.57	7.27	route it
R57	DQS4_T	245	NetR57_1	U5-C3	L1 + L3 or L1 + L8	5.79	7.91	14.71	in window
R58	DQS4_C	244	NetR58_1	U5-B3	L1 + L3 or L1 + L8	5.81	7.91	14.73	in window
R59	DM4	99	NetR59_1	U5-A7	L1 + L3 or L1 + L8	3.20	0.57	4.77	route it
R60	DQ37	240	NetR60_1	U5-D8	L1 + L3 or L1 + L8	5.70	0.57	7.27	route it

Byte 5 - U6 (target 12.65 mm, window 11.65 to 13.65)

Resistor	Finger net	J1 pin	DRAM-side net	DRAM ball	Layers	finger mm	DRAM mm	full mm	to do
R62	DQ40	108	NetR62_1	U6-C2	L1 + L3 or L1 + L8	3.14	0.57	4.70	route it
R63	DQ41	253	NetR63_1	U6-B7	L1 + L3 or L1 + L8	5.62	0.57	7.18	route it
R64	DQ42	115	NetR64_1	U6-D3	L1 + L3 or L1 + L8	2.99	0.57	4.56	route it
R65	DQ43	260	NetR65_1	U6-D7	L1 + L3 or L1 + L8	5.47	0.57	7.04	route it
R66	DQ44	106	NetR66_1	U6-D2	L1 + L3 or L1 + L8	3.18	0.57	4.74	route it
R67	DQ46	113	NetR67_1	U6-E3	L1 + L3 or L1 + L8	3.03	0.57	4.60	route it
R68	DQ47	258	NetR68_1	U6-E7	L1 + L3 or L1 + L8	5.51	0.57	7.08	route it
R69	DQS5_T	256	NetR69_1	U6-C3	L1 + L3 or L1 + L8	5.65	5.96	12.61	in window
R70	DQS5_C	255	NetR70_1	U6-B3	L1 + L3 or L1 + L8	5.69	5.96	12.65	in window
R71	DM5	110	NetR71_1	U6-A7	L1 + L3 or L1 + L8	3.10	0.57	4.66	route it
R72	DQ45	251	NetR72_1	U6-D8	L1 + L3 or L1 + L8	5.66	0.57	7.22	route it

Byte 6 - U7 (target 12.04 mm, window 11.04 to 13.04)

Resistor	Finger net	J1 pin	DRAM-side net	DRAM ball	Layers	finger mm	DRAM mm	full mm	to do
R74	DQ48	119	NetR74_1	U7-C2	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R75	DQ49	264	NetR75_1	U7-B7	L1 + L3 or L1 + L8	5.45	0.57	7.02	route it
R76	DQ50	126	NetR76_1	U7-D3	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R77	DQ51	271	NetR77_1	U7-D7	L1 + L3 or L1 + L8	5.45	0.57	7.02	route it
R78	DQ52	117	NetR78_1	U7-D2	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R79	DQ54	124	NetR79_1	U7-E3	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R80	DQ55	269	NetR80_1	U7-E7	L1 + L3 or L1 + L8	5.45	0.57	7.02	route it
R81	DQS6_T	267	NetR81_1	U7-C3	L1 + L3 or L1 + L8	5.54	5.50	12.04	in window
R82	DQS6_C	266	NetR82_1	U7-B3	L1 + L3 or L1 + L8	5.54	5.17	11.71	in window
R83	DM6	121	NetR83_1	U7-A7	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R84	DQ53	262	NetR84_1	U7-D8	L1 + L3 or L1 + L8	5.47	0.57	7.04	route it

Byte 7 - U8 (target 13.02 mm, window 12.02 to 14.02)

Resistor	Finger net	J1 pin	DRAM-side net	DRAM ball	Layers	finger mm	DRAM mm	full mm	to do
R86	DQ56	130	NetR86_1	U8-C2	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R87	DQ57	275	NetR87_1	U8-B7	L1 + L3 or L1 + L8	5.45	0.57	7.02	route it
R88	DQ58	137	NetR88_1	U8-D3	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R89	DQ59	282	NetR89_1	U8-D7	L1 + L3 or L1 + L8	5.45	0.57	7.02	route it
R90	DQ60	128	NetR90_1	U8-D2	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R91	DQ62	135	NetR91_1	U8-E3	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R92	DQ63	280	NetR92_1	U8-E7	L1 + L3 or L1 + L8	5.45	0.57	7.02	route it
R93	DQS7_T	278	NetR93_1	U8-C3	L1 + L3 or L1 + L8	5.56	6.45	13.02	in window
R94	DQS7_C	277	NetR94_1	U8-B3	L1 + L3 or L1 + L8	5.54	6.45	12.99	in window
R95	DM7	132	NetR95_1	U8-A7	L1 + L3 or L1 + L8	2.95	0.57	4.52	route it
R96	DQ61	273	NetR96_1	U8-D8	L1 + L3 or L1 + L8	5.45	0.57	7.02	route it

5. Address, command and control

Fly-by from J1 through U1 to U8 and then into the 39 ohm termination to VTT. Route on **L5** and **L6**, which are empty and reference VDD as Table 12 requires. Width 0.075 mm (55 ohm). Within 1.0 mm of each other and within CK +/- 0.5 mm.

Resistor	Signal	Function	Layers	Status
R97	A0	39 ohm to VTT	L5 / L6	done
R98	BG1	39 ohm to VTT	L5 / L6	done
R99	BG0	39 ohm to VTT	L5 / L6	done
R100	A1	39 ohm to VTT	L5 / L6	done
R101	CS	39 ohm to VTT	L5 / L6	done
R103	A2	39 ohm to VTT	L5 / L6	done
R104	CKE	39 ohm to VTT	L5 / L6	done
R105	A3	39 ohm to VTT	L5 / L6	done
R107	ODT	39 ohm to VTT	L5 / L6	done
R108	A4	39 ohm to VTT	L5 / L6	done
R109	WE	39 ohm to VTT	L5 / L6	done
R110	A5	39 ohm to VTT	L5 / L6	done
R111	A15_CAS	39 ohm to VTT	L5 / L6	done
R112	A6	39 ohm to VTT	L5 / L6	done
R113	A16_RAS	39 ohm to VTT	L5 / L6	done
R114	A7	39 ohm to VTT	L5 / L6	done
R116	A8	39 ohm to VTT	L5 / L6	done
R118	BA0	39 ohm to VTT	L5 / L6	done
R119	A9	39 ohm to VTT	L5 / L6	done
R120	BA1	39 ohm to VTT	L5 / L6	done
R121	A10_AP	39 ohm to VTT	L5 / L6	done
R122	A11	39 ohm to VTT	L5 / L6	done
R123	A12_BC	39 ohm to VTT	L5 / L6	done
R124	A13	39 ohm to VTT	L5 / L6	done
R125	ACT	39 ohm to VTT	L5 / L6	done
R126	PAR	39 ohm to VTT	L5 / L6	done

6. Clocks, ZQ, ALERT_n and SPD

Component	Nets	Function	Layers	Notes
R115	CK0_T / NetC56_1	39 ohm, clock termination after the last DRAM	L1, L6, L8	pairs with R117 and C56
R117	CK0_C / NetC56_1	39 ohm, clock termination	L1, L6, L8	pairs with R115
C56	NetC56_1 / VDD	0.01 uF from the termination node to VDD	-	main spec Table 9 note 2
R106	CK1_T / CK1_C	75 ohm across the unused clock pair	L1, L6, L8	terminate at the connector end, Annex A
R102	ALERT_n / VDD	47 ohm pull-up	L1, L3, L5, L6, L8	must sit BEFORE the first DRAM (Annex A 6.3.7)
R1	NetR1_1 / GND	240 ohm 1% ZQ for U1	L1	straight to ball B9, keep it short
R13	NetR13_1 / GND	240 ohm 1% ZQ for U2	L1	straight to ball B9, keep it short
R25	NetR25_1 / GND	240 ohm 1% ZQ for U3	L1	straight to ball B9, keep it short
R37	NetR37_1 / GND	240 ohm 1% ZQ for U4	L1	straight to ball B9, keep it short
R49	NetR49_1 / GND	240 ohm 1% ZQ for U5	L1	straight to ball B9, keep it short
R61	NetR61_1 / GND	240 ohm 1% ZQ for U6	L1	straight to ball B9, keep it short

R73	NetR73_1 / GND	240 ohm 1% ZQ for U7	L1	straight to ball B9, keep it short
R85	NetR85_1 / GND	240 ohm 1% ZQ for U8	L1	straight to ball B9, keep it short
U9	SA0, SA1, SA2, SCL, SDA	SPD EEPROM 34AA04	any signal layer	low speed, no length rules

7. Differential pairs

Pair	Positive	Negative	Where	Layers
CK0	CK0_T	CK0_C	connector side	L1, L6, L8
CK1	CK1_T	CK1_C	connector side	L1, L6, L8
DQS0	DQS0_T	DQS0_C	connector side	L1, L3, L8
DQS0_DRAM	NetR9_1	NetR10_1	DRAM side	L1, L3, L8
DQS1	DQS1_T	DQS1_C	connector side	L1, L3, L8
DQS1_DRAM	NetR21_1	NetR22_1	DRAM side	L1, L3, L8
DQS2	DQS2_T	DQS2_C	connector side	L1, L3, L8
DQS2_DRAM	NetR33_1	NetR34_1	DRAM side	L1, L3, L8
DQS3	DQS3_T	DQS3_C	connector side	L1, L3, L8
DQS3_DRAM	NetR45_1	NetR46_1	DRAM side	L1, L3, L8
DQS4	DQS4_T	DQS4_C	connector side	L1, L3, L8
DQS4_DRAM	NetR57_1	NetR58_1	DRAM side	L1, L3, L8
DQS5	DQS5_T	DQS5_C	connector side	L1, L3, L8
DQS5_DRAM	NetR69_1	NetR70_1	DRAM side	L1, L3, L8
DQS6	DQS6_T	DQS6_C	connector side	L1, L3, L8
DQS6_DRAM	NetR81_1	NetR82_1	DRAM side	L1, L3, L8
DQS7	DQS7_T	DQS7_C	connector side	L1, L3, L8
DQS7_DRAM	NetR93_1	NetR94_1	DRAM side	L1, L3, L8

Use **Route > Interactive Differential Pair Routing** (U, P) for these, never two single tracks. Halves matched within 0.1 mm; the pair as a whole is matched to its byte.

8. Special cases and things that catch people out

Differential pairs

18 pairs: CK0, CK1 and DQS0-DQS7 on the connector side, plus DQS0_DRAM-DQS7_DRAM between the 15 ohm resistors and the DRAMs. Route with Route > Interactive Differential Pair Routing (U, P) so both wires are drawn together and stay phase matched. Gap 0.10 mm, halves matched within 0.1 mm (main spec Table 11, rule MatchedLengths_DDR_CLOCKS_STROBES).

Byte-lane length matching

Each DQ and DM must sit within DQS +/- 1.0 mm over the whole finger-to-ball path, and the byte total must be 12.0-32.0 mm compensated (Table 11). The longest line in the byte is the reference: adding length to it lengthens every other line in that byte, so never tune the longest one.

xSignals

Once a byte is routed, Design > xSignals > Create xSignals with source J1 and destination Un traces the finger -> 15 ohm -> ball path as one object (4 nodes). Put the 11 into an xSignal class BYTEn_FULL and point a Matched Lengths rule at it with tolerance 1 mm. It fails on unrouted nets, so route first, then create.

Address fly-by

J1 -> U1 -> U2 -> U3 -> U4 -> U5 -> U6 -> U7 -> U8, then the 39 ohm termination to VTT. Address and control within 1.0 mm of each other and within CK +/- 0.5 mm; CK first-to-last DRAM at most 153 mm; the stub from the fly-by to each DRAM ball (TL2) at most 3.0 mm (Annex A).

Velocity compensation

All length rules are on stripline-equivalent lengths: divide any microstrip (L1 or L8) length by 1.1 before comparing (main spec 6.3.4). tools/check_lengths.py reports both.

Plane referencing

Table 12: DQ, DQS and DM reference GND; address, command, clock and VREFCA reference VDD. No signal may cross a plane split - check L2, L4 and L7 under every route.

Vias and fan-out

All vias are through-hole, 0.40 mm pad / 0.20 mm drill (0.45 / 0.25 for power). Routing space per Table 14: via-to-track 0.20 mm, large via 0.25 mm, track-to-pad 0.125 mm, track-to-shape 0.20 mm. All vias are tented (rule SolderMask_TentedVias).

Serpentine style

Accordion, Rounded, amplitude up to 1.0 mm in the corridor down the middle of a DRAM and 0.4 mm near the balls, spacing 0.3 mm = 3 x track width. Meander space: the corridor between ball columns 3 and 7 of each DRAM (about 3.2 x 9.6 mm) and the band just above the resistor rows.

9. What is left to route

Byte	DRAM side still to route	Routed but needs length tuning
Byte 0 (U1)	-	R2-R7, R9-R12
Byte 1 (U2)	R14-R20, R23	-
Byte 2 (U3)	R26-R32, R35-R36	R33
Byte 3 (U4)	R38-R44, R47-R48	-
Byte 4 (U5)	R50-R56, R59-R60	-
Byte 5 (U6)	R62-R68, R71-R72	-
Byte 6 (U7)	R74-R80, R83-R84	-
Byte 7 (U8)	R86-R92, R95-R96	-

Address and command: the fly-by from the first DRAM onwards is not routed yet - only the connector stubs exist. Plan it on L5 and L6. Clocks CK1_T / CK1_C are unrouted (terminate at the connector). SPD nets are unrouted.

10. Decoupling and bulk capacitors

Every capacitor sits between a supply and its return, so none of them is routed as a signal: connect each pad to the nearest plane with its own via, as short as possible.

Nets	Capacitors	Count
GND/VDD	C2-C4, C8-C10, C14-C16, C20-C22, C26-C28, C32-C34, C38-C40, C44-C46, C50-C52, C55	28
GND/VPP	C5-C6, C11-C12, C17-C18, C23-C24, C29-C30, C35-C36, C41-C42, C47-C48, C53	17
VDD/VTT	C57-C70	14
VDD/VREFCA	C1, C7, C13, C19, C25, C31, C37, C43, C54	9
NetC56_1/VDD	C56	1
GND/VDDSPD	C49	1

11. Full signal cross-reference

Every signal net on the board: where it starts at the connector, which resistor is in series or terminates it, which DRAM pins it lands on, and the layers it may use.

Net	J1 pin	Resistor	DRAM / SPD pin	Cap	Class	Layers	copper mm
A0	79	R97.1	U1-L3, U2-L3, U3-L3, U4-L3, U5-L3, U6-L3, U7-L3, U8-L3	-	Address	L1, L3, L5, L6, L8	4.53
A1	72	R100.1	U1-L7, U2-L7, U3-L7, U4-L7, U5-L7, U6-L7, U7-L7, U8-L7	-	Address	L1, L3, L5, L6, L8	4.53

A2	216	R103.1	U1-M3, U2-M3, U3-M3, U4-M3, U5-M3, U6-M3, U7-M3, U8-M3	-	Address	L1, L3, L5, L6, L8	4.53
A3	71	R105.1	U1-K7, U2-K7, U3-K7, U4-K7, U5-K7, U6-K7, U7-K7, U8-K7	-	Address	L1, L3, L5, L6, L8	4.53
A4	214	R108.1	U1-K3, U2-K3, U3-K3, U4-K3, U5-K3, U6-K3, U7-K3, U8-K3	-	Address	L1, L3, L5, L6, L8	4.53
A5	213	R110.1	U1-L8, U2-L8, U3-L8, U4-L8, U5-L8, U6-L8, U7-L8, U8-L8	-	Address	L1, L3, L5, L6, L8	4.53
A6	69	R112.1	U1-L2, U2-L2, U3-L2, U4-L2, U5-L2, U6-L2, U7-L2, U8-L2	-	Address	L1, L3, L5, L6, L8	4.53
A7	211	R114.1	U1-M8, U2-M8, U3-M8, U4-M8, U5-M8, U6-M8, U7-M8, U8-M8	-	Address	L1, L3, L5, L6, L8	4.53
A8	68	R116.1	U1-M2, U2-M2, U3-M2, U4-M2, U5-M2, U6-M2, U7-M2, U8-M2	-	Address	L1, L3, L5, L6, L8	4.53
A9	66	R119.1	U1-M7, U2-M7, U3-M7, U4-M7, U5-M7, U6-M7, U7-M7, U8-M7	-	Address	L1, L3, L5, L6, L8	4.53
A10_AP	225	R121.1	U1-J3, U2-J3, U3-J3, U4-J3, U5-J3, U6-J3, U7-J3, U8-J3	-	Address	L1, L3, L5, L6, L8	4.53
A11	210	R122.1	U1-N2, U2-N2, U3-N2, U4-N2, U5-N2, U6-N2, U7-N2, U8-N2	-	Address	L1, L3, L5, L6, L8	4.53
A12_BC	65	R123.1	U1-J7, U2-J7, U3-J7, U4-J7, U5-J7, U6-J7, U7-J7, U8-J7	-	Address	L1, L3, L5, L6, L8	4.53
A13	232	R124.1	U1-N8, U2-N8, U3-N8, U4-N8, U5-N8, U6-N8, U7-N8, U8-N8	-	Address	L1, L3, L5, L6, L8	4.53
A15_CAS	86	R111.1	U1-H7, U2-H7, U3-H7, U4-H7, U5-H7, U6-H7, U7-H7, U8-H7	-	Address	L1, L3, L5, L6, L8	4.53
A16_RAS	82	R113.1	U1-H8, U2-H8, U3-H8, U4-H8, U5-H8, U6-H8, U7-H8, U8-H8	-	Address	L1, L3, L5, L6, L8	4.53
ACT	62	R125.1	U1-H3, U2-H3, U3-H3, U4-H3, U5-H3, U6-H3, U7-H3, U8-H3	-	Address	L1, L3, L5, L6, L8	4.53
ALERT_n	208	R102.1	U1-L9, U2-L9, U3-L9, U4-L9, U5-L9, U6-L9, U7-L9, U8-L9	-	ALERT	L1, L3, L5, L6, L8	4.53
BA0	81	R118.1	U1-K2, U2-K2, U3-K2, U4-K2, U5-K2, U6-K2, U7-K2, U8-K2	-	Address	L1, L3, L5, L6, L8	4.53
BA1	224	R120.1	U1-K8, U2-K8, U3-K8, U4-K8, U5-K8, U6-K8, U7-K8, U8-K8	-	Address	L1, L3, L5, L6, L8	4.53
BG0	63	R99.1	U1-J2, U2-J2, U3-J2, U4-J2, U5-J2, U6-J2, U7-J2, U8-J2	-	Address	L1, L3, L5, L6, L8	4.53
BG1	207	R98.1	U1-J8, U2-J8, U3-J8, U4-J8, U5-J8, U6-J8, U7-J8, U8-J8	-	Address	L1, L3, L5, L6, L8	4.53
CK0_C	75	R117.1	U1-F8, U2-F8, U3-F8, U4-F8, U5-F8, U6-F8, U7-F8, U8-F8	-	Clock	L1, L6, L8	4.53
CK0_T	74	R115.1	U1-F7, U2-F7, U3-F7, U4-F7, U5-F7, U6-F7, U7-F7, U8-F7	-	Clock	L1, L6, L8	4.53
CK1_C	219	R106.2	-	-	Clock	L1, L6, L8	0.00
CK1_T	218	R106.1	-	-	Clock	L1, L6, L8	0.00
CKE	60	R104.1	U1-G3, U2-G3, U3-G3, U4-G3, U5-G3, U6-G3, U7-G3, U8-G3	-	Control	L1, L3, L5, L6, L8	4.53
CS	84	R101.1	U1-G7, U2-G7, U3-G7, U4-G7, U5-G7, U6-G7, U7-G7, U8-G7	-	Control	L1, L3, L5, L6, L8	4.53
DM0	7	R11.2	-	-	Data	L1, L3, L8	2.95
DM1	18	R23.2	-	-	Data	L1, L3, L8	2.95
DM2	29	R35.2	-	-	Data	L1, L3, L8	2.95

DQ40	108	R62.2	-	-	Data	L1, L3, L8	3.14
DQ41	253	R63.2	-	-	Data	L1, L3, L8	5.62
DQ42	115	R64.2	-	-	Data	L1, L3, L8	2.99
DQ43	260	R65.2	-	-	Data	L1, L3, L8	5.47
DQ44	106	R66.2	-	-	Data	L1, L3, L8	3.18
DQ45	251	R72.2	-	-	Data	L1, L3, L8	5.66
DQ46	113	R67.2	-	-	Data	L1, L3, L8	3.03
DQ47	258	R68.2	-	-	Data	L1, L3, L8	5.51
DQ48	119	R74.2	-	-	Data	L1, L3, L8	2.95
DQ49	264	R75.2	-	-	Data	L1, L3, L8	5.45
DQ50	126	R76.2	-	-	Data	L1, L3, L8	2.95
DQ51	271	R77.2	-	-	Data	L1, L3, L8	5.45
DQ52	117	R78.2	-	-	Data	L1, L3, L8	2.95
DQ53	262	R84.2	-	-	Data	L1, L3, L8	5.47
DQ54	124	R79.2	-	-	Data	L1, L3, L8	2.95
DQ55	269	R80.2	-	-	Data	L1, L3, L8	5.45
DQ56	130	R86.2	-	-	Data	L1, L3, L8	2.95
DQ57	275	R87.2	-	-	Data	L1, L3, L8	5.45
DQ58	137	R88.2	-	-	Data	L1, L3, L8	2.95
DQ59	282	R89.2	-	-	Data	L1, L3, L8	5.45
DQ60	128	R90.2	-	-	Data	L1, L3, L8	2.95
DQ61	273	R96.2	-	-	Data	L1, L3, L8	5.45
DQ62	135	R91.2	-	-	Data	L1, L3, L8	2.95
DQ63	280	R92.2	-	-	Data	L1, L3, L8	5.45
DQS0_C	152	R10.2	-	-	Data	L1, L3, L8	5.54
DQS0_T	153	R9.2	-	-	Data	L1, L3, L8	5.54
DQS1_C	163	R22.2	-	-	Data	L1, L3, L8	5.54
DQS1_T	164	R21.2	-	-	Data	L1, L3, L8	5.54
DQS2_C	174	R34.2	-	-	Data	L1, L3, L8	5.56
DQS2_T	175	R33.2	-	-	Data	L1, L3, L8	5.56
DQS3_C	185	R46.2	-	-	Data	L1, L3, L8	5.67
DQS3_T	186	R45.2	-	-	Data	L1, L3, L8	5.71
DQS4_C	244	R58.2	-	-	Data	L1, L3, L8	5.81
DQS4_T	245	R57.2	-	-	Data	L1, L3, L8	5.79
DQS5_C	255	R70.2	-	-	Data	L1, L3, L8	5.69
DQS5_T	256	R69.2	-	-	Data	L1, L3, L8	5.65
DQS6_C	266	R82.2	-	-	Data	L1, L3, L8	5.54
DQS6_T	267	R81.2	-	-	Data	L1, L3, L8	5.54
DQS7_C	277	R94.2	-	-	Data	L1, L3, L8	5.54
DQS7_T	278	R93.2	-	-	Data	L1, L3, L8	5.56
NetC56_1	-	R115.2, R117.2	-	C56	-	L1, L3, L5, L6, L8	0.00
NetR1_1	-	R1.1	U1-B9	-	Data	L1, L3, L8	0.57

NetR2_1	-	R2.1	U1-C2	-	Data	L1, L3, L8	9.43
NetR3_1	-	R3.1	U1-B7	-	Data	L1, L3, L8	4.00
NetR4_1	-	R4.1	U1-D3	-	Data	L1, L3, L8	8.27
NetR5_1	-	R5.1	U1-D7	-	Data	L1, L3, L8	11.21
NetR6_1	-	R6.1	U1-D2	-	Data	L1, L3, L8	11.02
NetR7_1	-	R7.1	U1-E3	-	Data	L1, L3, L8	12.06
NetR8_1	-	R8.1	U1-E7	-	Data	L1, L3, L8	13.11
NetR9_1	-	R9.1	U1-C3	-	Data	L1, L3, L8	7.01
NetR10_1	-	R10.1	U1-B3	-	Data	L1, L3, L8	6.68
NetR11_1	-	R11.1	U1-A7	-	Data	L1, L3, L8	6.00
NetR12_1	-	R12.1	U1-D8	-	Data	L1, L3, L8	5.92
NetR13_1	-	R13.1	U2-B9	-	Data	L1, L3, L8	0.57
NetR14_1	-	R14.1	U2-C2	-	Data	L1, L3, L8	0.57
NetR15_1	-	R15.1	U2-B7	-	Data	L1, L3, L8	0.74
NetR16_1	-	R16.1	U2-D3	-	Data	L1, L3, L8	0.57
NetR17_1	-	R17.1	U2-D7	-	Data	L1, L3, L8	0.57
NetR18_1	-	R18.1	U2-D2	-	Data	L1, L3, L8	0.57
NetR19_1	-	R19.1	U2-E3	-	Data	L1, L3, L8	0.57
NetR20_1	-	R20.1	U2-E7	-	Data	L1, L3, L8	0.57
NetR21_1	-	R21.1	U2-C3	-	Data	L1, L3, L8	8.24
NetR22_1	-	R22.1	U2-B3	-	Data	L1, L3, L8	8.24
NetR23_1	-	R23.1	U2-A7	-	Data	L1, L3, L8	0.57
NetR24_1	-	R24.1	U2-D8	-	Data	L1, L3, L8	7.95
NetR25_1	-	R25.1	U3-B9	-	Data	L1, L3, L8	0.57
NetR26_1	-	R26.1	U3-C2	-	Data	L1, L3, L8	0.57
NetR27_1	-	R27.1	U3-B7	-	Data	L1, L3, L8	0.57
NetR28_1	-	R28.1	U3-D3	-	Data	L1, L3, L8	0.57
NetR29_1	-	R29.1	U3-D7	-	Data	L1, L3, L8	0.57
NetR30_1	-	R30.1	U3-D2	-	Data	L1, L3, L8	0.57
NetR31_1	-	R31.1	U3-E3	-	Data	L1, L3, L8	0.57
NetR32_1	-	R32.1	U3-E7	-	Data	L1, L3, L8	0.57
NetR33_1	-	R33.1	U3-C3	-	Data	L1, L3, L8	8.12
NetR34_1	-	R34.1	U3-B3	-	Data	L1, L3, L8	9.87
NetR35_1	-	R35.1	U3-A7	-	Data	L1, L3, L8	0.57
NetR36_1	-	R36.1	U3-D8	-	Data	L1, L3, L8	0.57
NetR37_1	-	R37.1	U4-B9	-	Data	L1, L3, L8	0.57
NetR38_1	-	R38.1	U4-C2	-	Data	L1, L3, L8	0.57
NetR39_1	-	R39.1	U4-B7	-	Data	L1, L3, L8	0.57
NetR40_1	-	R40.1	U4-D3	-	Data	L1, L3, L8	0.57
NetR41_1	-	R41.1	U4-D7	-	Data	L1, L3, L8	0.57
NetR42_1	-	R42.1	U4-D2	-	Data	L1, L3, L8	0.57
NetR43_1	-	R43.1	U4-E3	-	Data	L1, L3, L8	0.57

NetR44_1	-	R44.1	U4-E7	-	Data	L1, L3, L8	0.57
NetR45_1	-	R45.1	U4-C3	-	Data	L1, L3, L8	10.62
NetR46_1	-	R46.1	U4-B3	-	Data	L1, L3, L8	10.62
NetR47_1	-	R47.1	U4-A7	-	Data	L1, L3, L8	0.57
NetR48_1	-	R48.1	U4-D8	-	Data	L1, L3, L8	0.57
NetR49_1	-	R49.1	U5-B9	-	Data	L1, L3, L8	0.57
NetR50_1	-	R50.1	U5-C2	-	Data	L1, L3, L8	0.57
NetR51_1	-	R51.1	U5-B7	-	Data	L1, L3, L8	0.57
NetR52_1	-	R52.1	U5-D3	-	Data	L1, L3, L8	0.57
NetR53_1	-	R53.1	U5-D7	-	Data	L1, L3, L8	0.57
NetR54_1	-	R54.1	U5-D2	-	Data	L1, L3, L8	0.57
NetR55_1	-	R55.1	U5-E3	-	Data	L1, L3, L8	0.57
NetR56_1	-	R56.1	U5-E7	-	Data	L1, L3, L8	0.57
NetR57_1	-	R57.1	U5-C3	-	Data	L1, L3, L8	7.91
NetR58_1	-	R58.1	U5-B3	-	Data	L1, L3, L8	7.91
NetR59_1	-	R59.1	U5-A7	-	Data	L1, L3, L8	0.57
NetR60_1	-	R60.1	U5-D8	-	Data	L1, L3, L8	0.57
NetR61_1	-	R61.1	U6-B9	-	Data	L1, L3, L8	0.57
NetR62_1	-	R62.1	U6-C2	-	Data	L1, L3, L8	0.57
NetR63_1	-	R63.1	U6-B7	-	Data	L1, L3, L8	0.57
NetR64_1	-	R64.1	U6-D3	-	Data	L1, L3, L8	0.57
NetR65_1	-	R65.1	U6-D7	-	Data	L1, L3, L8	0.57
NetR66_1	-	R66.1	U6-D2	-	Data	L1, L3, L8	0.57
NetR67_1	-	R67.1	U6-E3	-	Data	L1, L3, L8	0.57
NetR68_1	-	R68.1	U6-E7	-	Data	L1, L3, L8	0.57
NetR69_1	-	R69.1	U6-C3	-	Data	L1, L3, L8	5.96
NetR70_1	-	R70.1	U6-B3	-	Data	L1, L3, L8	5.96
NetR71_1	-	R71.1	U6-A7	-	Data	L1, L3, L8	0.57
NetR72_1	-	R72.1	U6-D8	-	Data	L1, L3, L8	0.57
NetR73_1	-	R73.1	U7-B9	-	Data	L1, L3, L8	0.57
NetR74_1	-	R74.1	U7-C2	-	Data	L1, L3, L8	0.57
NetR75_1	-	R75.1	U7-B7	-	Data	L1, L3, L8	0.57
NetR76_1	-	R76.1	U7-D3	-	Data	L1, L3, L8	0.57
NetR77_1	-	R77.1	U7-D7	-	Data	L1, L3, L8	0.57
NetR78_1	-	R78.1	U7-D2	-	Data	L1, L3, L8	0.57
NetR79_1	-	R79.1	U7-E3	-	Data	L1, L3, L8	0.57
NetR80_1	-	R80.1	U7-E7	-	Data	L1, L3, L8	0.57
NetR81_1	-	R81.1	U7-C3	-	Data	L1, L3, L8	5.50
NetR82_1	-	R82.1	U7-B3	-	Data	L1, L3, L8	5.17
NetR83_1	-	R83.1	U7-A7	-	Data	L1, L3, L8	0.57
NetR84_1	-	R84.1	U7-D8	-	Data	L1, L3, L8	0.57
NetR85_1	-	R85.1	U8-B9	-	Data	L1, L3, L8	0.57

NetR86_1	-	R86.1	U8-C2	-	Data	L1, L3, L8	0.57
NetR87_1	-	R87.1	U8-B7	-	Data	L1, L3, L8	0.57
NetR88_1	-	R88.1	U8-D3	-	Data	L1, L3, L8	0.57
NetR89_1	-	R89.1	U8-D7	-	Data	L1, L3, L8	0.57
NetR90_1	-	R90.1	U8-D2	-	Data	L1, L3, L8	0.57
NetR91_1	-	R91.1	U8-E3	-	Data	L1, L3, L8	0.57
NetR92_1	-	R92.1	U8-E7	-	Data	L1, L3, L8	0.57
NetR93_1	-	R93.1	U8-C3	-	Data	L1, L3, L8	6.45
NetR94_1	-	R94.1	U8-B3	-	Data	L1, L3, L8	6.45
NetR95_1	-	R95.1	U8-A7	-	Data	L1, L3, L8	0.57
NetR96_1	-	R96.1	U8-D8	-	Data	L1, L3, L8	0.57
ODT	87	R107.1	U1-F3, U2-F3, U3-F3, U4-F3, U5-F3, U6-F3, U7-F3, U8-F3	-	Control	L1, L3, L5, L6, L8	4.53
PAR	222	R126.1	U1-N3, U2-N3, U3-N3, U4-N3, U5-N3, U6-N3, U7-N3, U8-N3	-	Address	L1, L3, L5, L6, L8	4.53
RESET_N	58	-	U1-L1, U2-L1, U3-L1, U4-L1, U5-L1, U6-L1, U7-L1, U8-L1	-	Control	L1, L3, L5, L6, L8	4.53
SA0	139	-	U9-1	-	SPD	any signal layer	0.00
SA1	140	-	U9-2	-	SPD	any signal layer	0.00
SA2	238	-	U9-3	-	SPD	any signal layer	0.00
SCL	141	-	U9-6	-	SPD	any signal layer	0.00
SDA	285	-	U9-5	-	SPD	any signal layer	0.00
WE	228	R109.1	U1-H2, U2-H2, U3-H2, U4-H2, U5-H2, U6-H2, U7-H2, U8-H2	-	Address	L1, L3, L5, L6, L8	4.53